



Adding Connectivity to RTOS MCUs

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Agenda

1	Edge AI meets connectivity	3
2	Dual core support with Virtual Connectivity Manager (VCM)	14
	Lab 1: Machine Learning Demo Using Dual Core Virtual MQTT Client	
3	Efficient Bluetooth® LE audio	15
	Lab 2: Bluetooth® LE Audio leveraging Helium extension	
4	Low-power assistant (LPA)	19
	Lab 3: WLAN Low-Power Assistant	
5	Other key advantages	21

Edge AI meets connectivity



Connected Edge AI Benefits

Real-Time Processing & Wireless Connectivity

Instant decision-making and seamless cloud communication

Lower Latency & Faster Response

Reduces cloud reliance while enabling quick local data exchange

Reduced Bandwidth & Cloud Costs

Minimizes the need to send large data amounts to the cloud

Improved Security & Privacy

Local data processing reduces exposure to cyber threats

Energy Efficiency

Intelligent power management optimizes wireless usage

Scalability & Flexibility

Independent operations reduce network congestions

By combining Edge AI with Wi-Fi and Bluetooth, industries unlock faster, smarter and more efficient automation,
Enhancing everything from factories to smart homes

PSOC™ Edge E8 MCU family overview

	PSOC™ Edge E81 Base	PSOC™ Edge E82 Base + GFX	PSOC™ Edge E83 Base + NPU + Vision	PSOC™ Edge E84 Base + NPU + Vision + GFX
Processor	Cortex®-M55 + DSP Cortex®-M33	Cortex®-M55 + DSP Cortex®-M33	Cortex®-M55 + DSP Cortex®-M33	Cortex®-M55 + DSP Cortex®-M33
Machine Learning	DSP NN accelerator, NNLite	DSP NN accelerator, NNLite	DSP NN accelerator, NNLite, U55-128	DSP NN accelerator, NNLite, U55-128
SRAM	Up to 4 MB (System RAM) Up to 1 MB (Low-power domain)	Up to 4 MB (System RAM) Up to 1 MB (Low-power domain)	Up to 4 MB (System RAM) Up to 1 MB (low-power domain)	Up to 5 MB (System RAM) Up to 1 MB (low-power domain)
RRAM / NVM	512 kB	512 kB	512 kB	512 kB
External Memory	2x SMIF, 2x SD Host Controller	2x SMIF, 2x SD Host Controller	2x SMIF, 2x SD Host Controller	2x SMIF, 2x SD Host Controller
Audio/Voice	ULP Always ON prog. analog for voice, audio, sensing 4x Analog mic, 6x Digital mic Acoustic activity detection & NNLite Wake Word	ULP Always ON prog. analog for voice, audio, sensing 4x Analog mic, 6x Digital mic Acoustic activity detection & NNLite Wake Word	ULP Always ON prog. analog for voice, audio, sensing 4x Analog mic, 6x Digital mic Acoustic activity detection & U55 ML-based Wake Word & Full voice inferencing	ULP Always ON prog. analog for voice, audio, sensing 4x Analog Mic, 6x Digital mic Acoustic activity detection & U55 ML-based Wake Word, & Full voice inferencing
Graphics	No	LP 2.5D GPU Up to 1024 x 768, MIPI-DSI/DBI formats	No	LP 2.5D GPU Up to 1024 x 768, MIPI-DSI/DBI formats
Vision	No	No	Position detection/Face recognition/Object detection (VGA) via USB	Position detection/Face recognition/Object detection (VGA) via USB
Peripherals & IO	USB, 10/100 Ethernet, CAN, SPI, UART, I2C, I3C, I2S	USB, 10/100 Ethernet, CAN, SPI, UART, I2C, I3C, I2S	USB, 10/100 Ethernet, CAN, SPI, UART, I2C, I3C, I2S	USB, 10/100 Ethernet, CAN, SPI, UART, I2C, I3C, I2S
Security	Infineon EPC4 with PSA Certified L3/L4 iSE Edge-Protect Bootloader and TF-M stack	Infineon EPC4 with PSA Certified L3/L4 iSE Edge-Protect Bootloader and TF-M stack	Infineon EPC4 with PSA Certified L3/L4 iSE Edge-Protect Bootloader and TF-M stack	Infineon EPC4 with PSA Certified L3/L4 iSE Edge-Protect Bootloader and TF-M stack
Packages	BGA-220, 10x10, 0.65p WLB-154, 4.3x5.3, 0.35p	BGA-220, 10x10, 0.65p WLB-154, 4.3x5.3, 0.35p	BGA-220, 10x10, 0.65p WLB-154, 4.3x5.3, 0.35p	BGA-220, 10x10, 0.65p eWLB-235, 7x7, 0.4p WLB-154, 4.3x5.3, 0.35p
Availability	Available now	Available now	Available now	Available now

AIROC™ Wi-Fi Combo portfolio addresses a wide range of applications



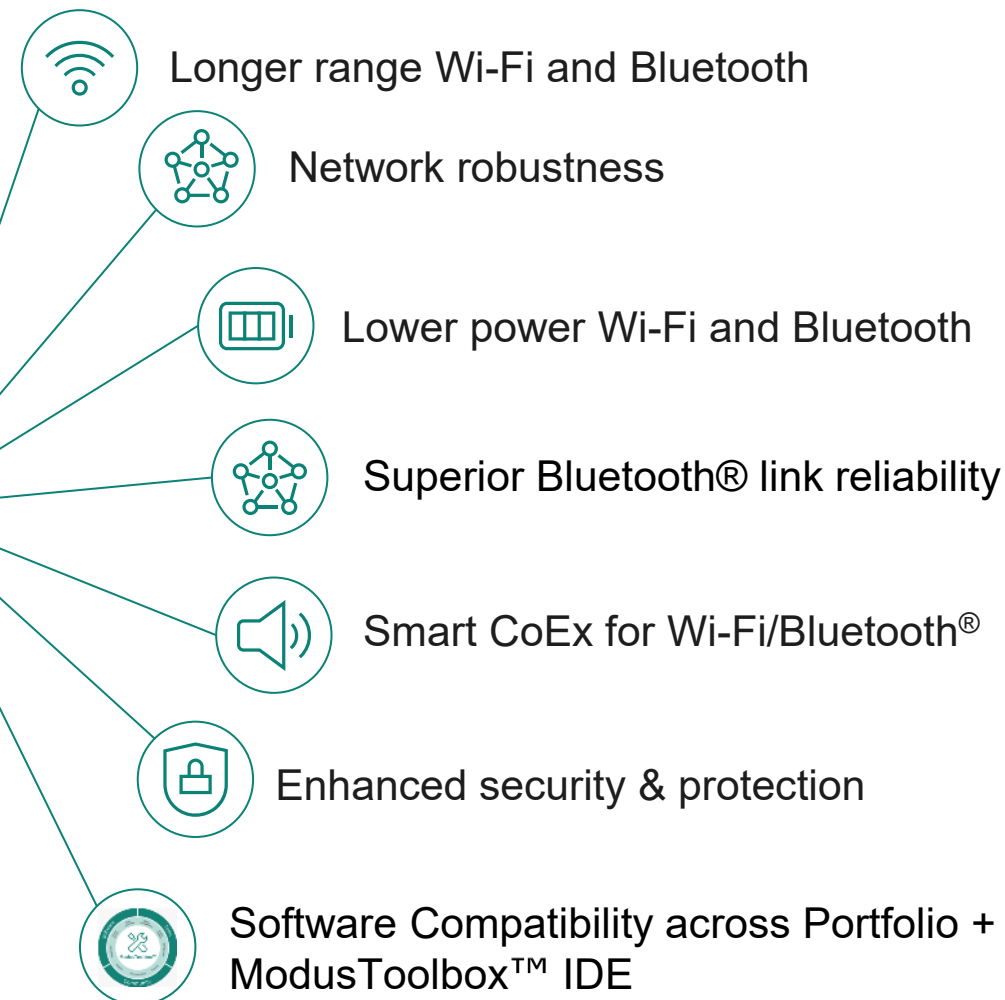
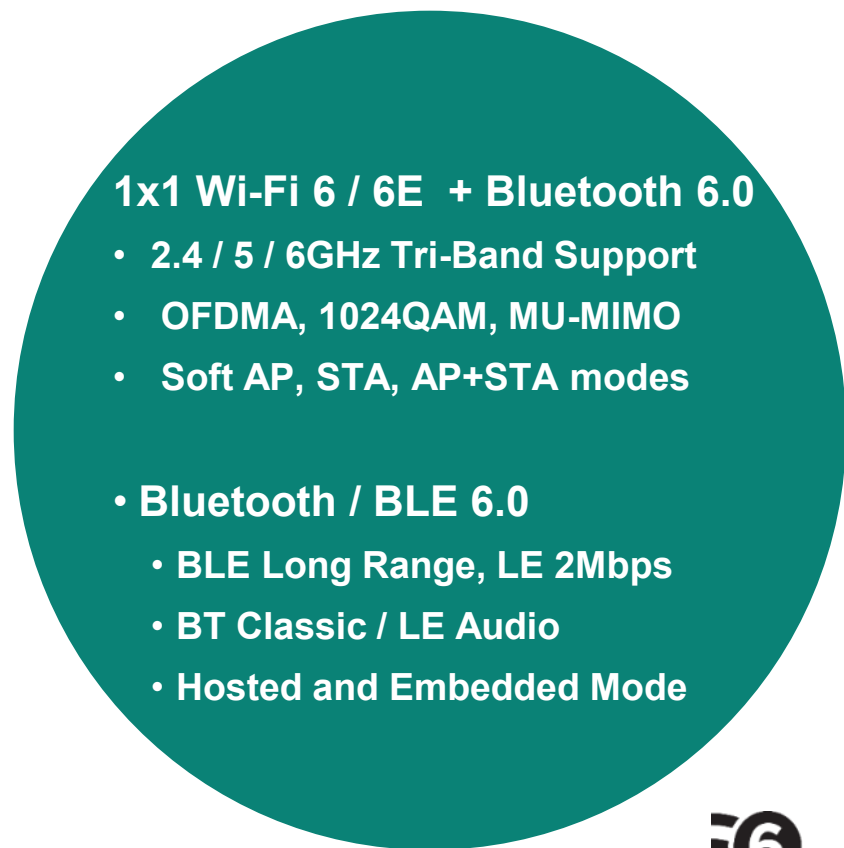
	4 5	6 6E		Values	Application examples
2x2 80MHz DB/TB Wi-Fi Combo	 5459x	 55573/2	High Performance	High data rate (>300 Mbps), MIMO 2-Antenna, Longer range, Highest reliability	Video conferencing, High-end security camera, Industrial, EV Charging system, Commercial scanner/printer, Automotive Infotainment etc.
1x1 80MHz DB/TB Wi-Fi Combo	 4373	 55571/0		Medium data rate (100Mbps+), Single antenna, Balance between performance, reliability, form factor, power	Smart speaker, Video streaming, Digital camera, Action camera, Drone, AR glasses, Industrial, Healthcare/Medical etc.
1x1 20MHz DB/TB Wi-Fi Combo	 43022	 55512/3	Low Power	Low data rate (50Mbps-100Mbps, typically 5-10Mbps or less for IoT), Low power, Reliable and long range	Security camera, Doorbell, Printer, Smart lock, Home appliances, Thermostat, Industrial, Healthcare/Medical, Wearable etc.
1x1 20MHz SB Wi-Fi Combo	 43439	 55511	Entry-level		
	 55532/3	 55531		Low data rate, 2.4 GHz only, Lowest cost Wi-Fi option	Low-end Security Camera, Printer, Home appliances etc.

SB: Single-band, DB: Dual-band, TB: Tri-band

AIROC™ CYW5551x Wi-Fi 6E + Bluetooth 6.0 Combo

CYW5551x Family

CYW5551x Benefits



Applications Examples (but not limited to...)

- The PSOC™ Edge family excels at enabling AI/ML at the Edge with ultra-low power consumption, high performance, real-time processing capabilities and industry-leading security
- The CYW5551x AIROC™ family excels at applications that require best in class RF performance for Wi-Fi and Bluetooth®, battery powered/power sensitive Wi-Fi/Bluetooth® applications and applications requiring Bluetooth®/BLE Audio support



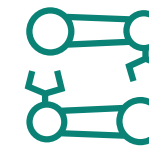
Smart Buildings



Surveillance Camera
Video Doorbell



Smart Homes



Industrial



Gateways
Bridges/Hubs



Wearables



Appliances



Healthcare

Use case: Fleet Management for AMRs



Challenges:

- Close to zero latency requirement for local decisions (safety)
- Robots operate in large warehouses
- Communications go through obstacles causing weak signals
- Robots lose connections/ communications

Connected Edge AI benefits:

- Autonomous operation / obstacle detection at the edge
- Cloud connection for warehouse mapping and route optimization

Vision Comparison	Inference Time	Energy	Cost
CPU @ 2.4GHz			\$\$\$
Arm Cortex-M7 @ 1.0GHz			\$\$
PSOC™ Edge E84 Cortex-M55 & Ethos-U55 @ 400MHz			\$

- **Performance:** Approaching GHz CPU/MPU capability
- **Power:** > 200x energy savings
- **Cost:** ½ total system cost



>40% longer range and better coverage



Wi-Fi: SDIO

BT/BLE: UART

Control Signals: wake-up, enable, etc...



PSOC™ Edge E84 AI kit: KIT_PSE84_AI

Kit contents

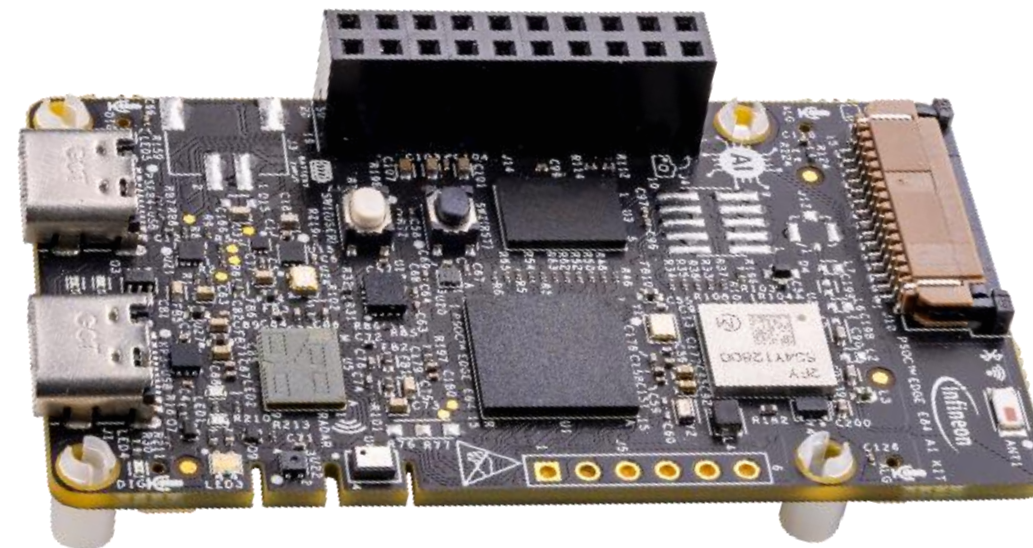
- PSOC™ Edge E84 AI kit
- Camera module

Highlights

- Designed to support the creation of Edge AI-powered applications
- Optimized for rapid prototyping and development of embedded systems leveraging the versatile PSOC™ Edge E84 microcontroller
- Data collection enabled via radar, acoustic, pressure, and IMU sensors available in the kit
- Enables evaluation of Infineon's ML platform – DEEPCRAFT™ AI Suite

Featured components

- PSE846GPS2DBZC4 PSOC™ Edge E84 MCU
- AIROC™ CY55513 Wi-Fi + Bluetooth® combo chip
- BGT60TR13C 60 GHz radar
- IM73A135V01 analog microphone
- IM73D122V01 digital microphone
- DPS368 barometric pressure sensor
- S25FS128SAGBHM203 16 MB flash memory
- S70KS1283GABHI020 16 MB PSRAM



Kit status

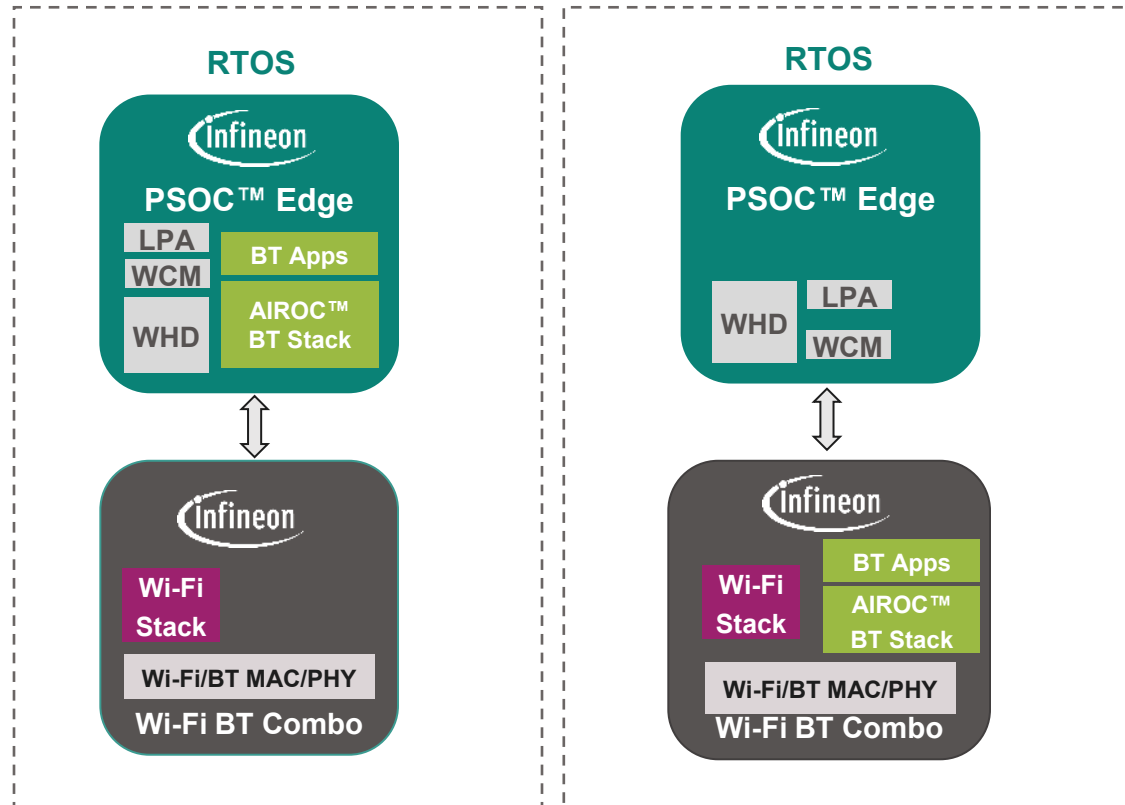
- **Available now**
- Order by visiting [KIT_PSE84_AI](#)

Software Integration

PSOC™ Edge + AIROC™ CYW5551x Connectivity

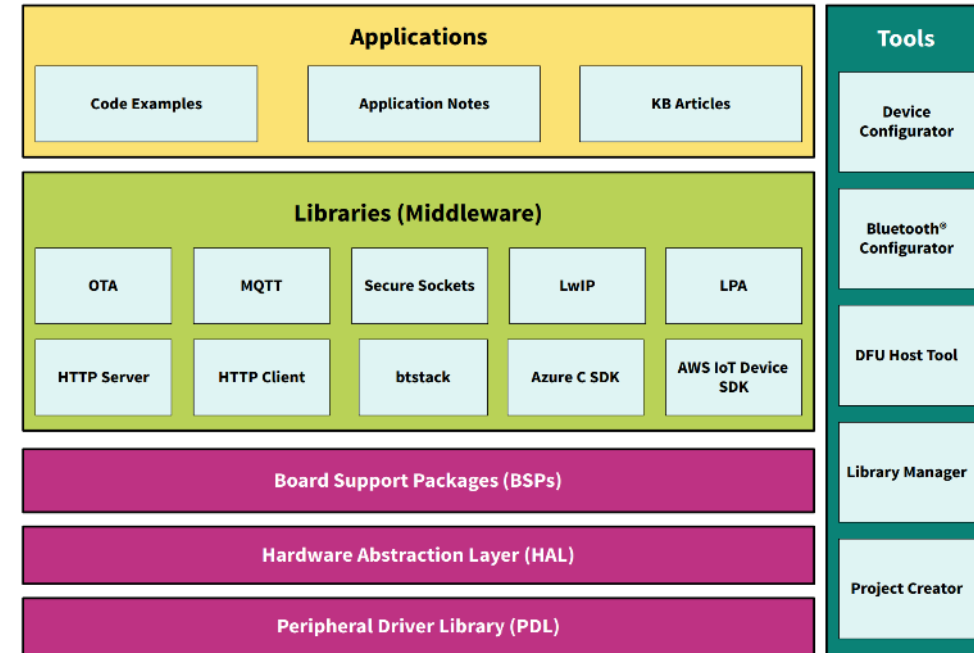
Wi-Fi + Bluetooth® HCI mode

Wi-Fi + Bluetooth® Embedded mode



WHD = Wireless Host Driver, HCI = Host Controller Interface
WCM = Wireless Connection Manager
LPA = Low Power Assistant

Connectivity Solution in ModusToolbox™



Examples of software modules for Wi-Fi and BT

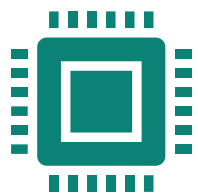
Wi-Fi

- ☐ PSOC Edge Azure IOT
- ☐ PSOC Edge Bluetooth LE W-Fi IoT gateway
- ☐ PSOC Edge Tester - Enterprise security
- ☐ PSOC Edge Tester - Wi-Fi Bluetooth Console
- ☐ PSOC Edge Wi-Fi dual-core virtual MQTT client
- ☐ PSOC Edge Wi-Fi HTTPS Client
- ☐ PSOC Edge Wi-Fi HTTPS Server
- ☐ PSOC Edge Wi-Fi MQTT Client
- ☐ PSOC Edge Wi-Fi Scan
- ☐ PSOC Edge Wi-Fi Secure TCP Client
- ☐ PSOC Edge Wi-Fi Secure TCP Server
- ☐ PSOC Edge Wi-Fi TCP Client
- ☐ PSOC Edge Wi-Fi TCP Server
- ☐ PSOC Edge Wi-Fi UDP Client
- ☐ PSOC Edge Wi-Fi UDP Server
- ☐ PSOC Edge Wi-Fi Web Server
- ☐ PSOC Edge WLAN Low Power
- ☐ PSOC Edge WLAN Offloads

Bluetooth®

- ☐ PSOC Edge Bluetooth A2DP Sink
- ☐ PSOC Edge Bluetooth A2DP Source
- ☐ PSOC Edge Bluetooth EATT Client
- ☐ PSOC Edge Bluetooth EATT Server
- ☐ PSOC Edge Bluetooth HFP Hands Free
- ☐ PSOC Edge Bluetooth LE Alert Client
- ☐ PSOC Edge Bluetooth LE Alert Server
- ☐ PSOC Edge Bluetooth LE CTS Client
- ☐ PSOC Edge Bluetooth LE CTS Server
- ☐ PSOC Edge Bluetooth LE Environmental Sensing Service
- ☐ PSOC Edge Bluetooth LE FindMe
- ☐ PSOC Edge Bluetooth LE GATT Caching Client
- ☐ PSOC Edge Bluetooth LE GATT Caching Server
- ☐ PSOC Edge Bluetooth LE Hello Sensor
- ☐ PSOC Edge Bluetooth LE HFP Audio Gateway
- ☐ PSOC Edge Bluetooth LE ISOC Central
- ☐ PSOC Edge Bluetooth LE ISOC Peripheral
- ☐ PSOC Edge Bluetooth LE Multi Beacon
- ☐ PSOC Edge Bluetooth LE Periodic Advertisement
- ☐ PSOC Edge Bluetooth LE Periodic Sync
- ☐ PSOC Edge Bluetooth LE Peripheral Privacy
- ☐ PSOC Edge Bluetooth LE Sensorhub
- ☐ PSOC Edge Bluetooth LE Unicast Sink
- ☐ PSOC Edge Bluetooth LE with CAPSENSE Button and Slider
- ☐ PSOC Edge Wi-Fi Onboarding Using Bluetooth LE

Success Story 1 – Industrial Forklift Display



- Fork Truck Operator Display:
PSOC™ Edge E8x + CYW5551x



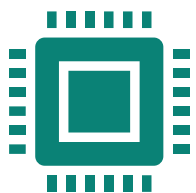
- Support for 105C
- GUI Design for display creation
- Long Range & Robust Connectivity



- Seamless connectivity integration makes for a full turnkey solution
- Peripheral set including USB and 2.5D GPU offers an integrated single chip solution
- Edge AI/ML capabilities allow for future product upgrades including voice and facial ID
- Leading security level (PSA L4) addresses security concerns



Success Story 2 – Wearable for Elderly People with Fall detection



- Wearable for Elderly people:
PSOC™ Edge E8x + CYW555xx



- Low Power
- Smart Fall detection
- Small Form factor
- Reliable connectivity



- Seamless connectivity integration makes for a full turnkey solution
- Dynamic and static ultra low power numbers offer best-in-class battery life
- Edge AI/ML capabilities allow for advanced fall detection model implementation
- Highest level of security addresses privacy concerns

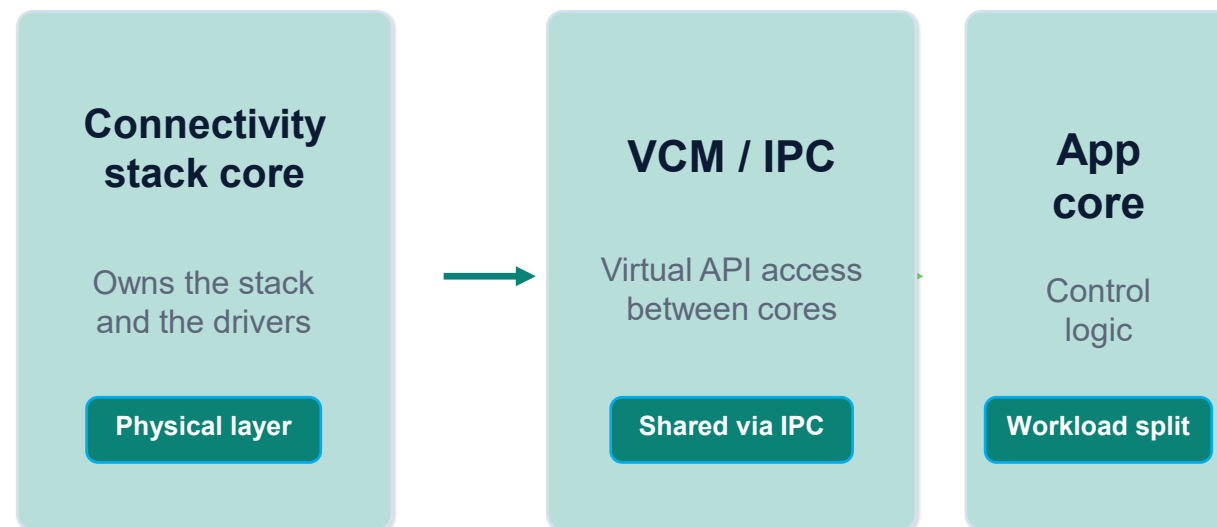


Dual core support with Virtual Connectivity Manager (VCM)

How VCM enables dual-core connectivity

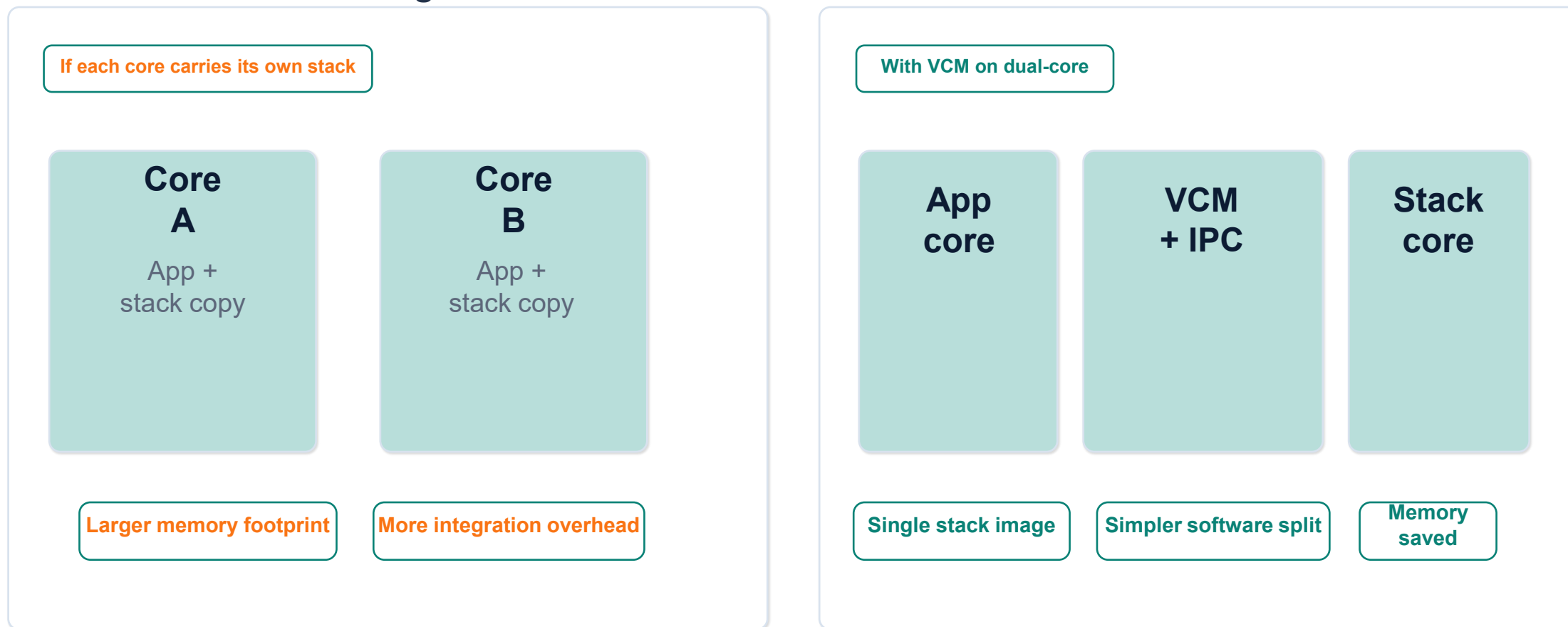
VCM virtualizes access to a connectivity stack on one core so the other core can still call the APIs through IPC.

- › One core owns the physical link and middleware stack.
- › The second core uses the same stack through a virtual interface.
- › This keeps a multi-core design looking like one connected system.



Shared stack instead of duplicated integration

Infineon's dual-core guidance highlights a practical advantage of VCM: share the connectivity libraries across cores instead of linking them into both binaries.



Partition for throughput or for low-power

Stack placement changes the trade-off, while VCM lets the other core keep using the stack.

Performance-leaning placement

Higher throughput on CM55 vs. CM33



When bandwidth matters, put the heavier networking work on the faster core.

Low-power leaning placement

Lower current consumption on CM33 vs. CM55



When energy is tighter, keep the stack on the lower-power core and let the other core access it virtually.

VCM turns this into a software partitioning choice — not a rewrite.

- Tune core placement for the product target.
- Keep application code on the other core.
- Scale without duplicating the connectivity stack.

Performance ↔ Power

Lab 1: Machine Learning Demo Using Dual Core Virtual MQTT Client

What will you learn?

- How to add connectivity to a machine learning demo on PSOC™ Edge
- How to add VCM to an application

Lab 1:

Machine Learning Demo Using Dual Core Virtual MQTT Client



Follow steps in training manual:

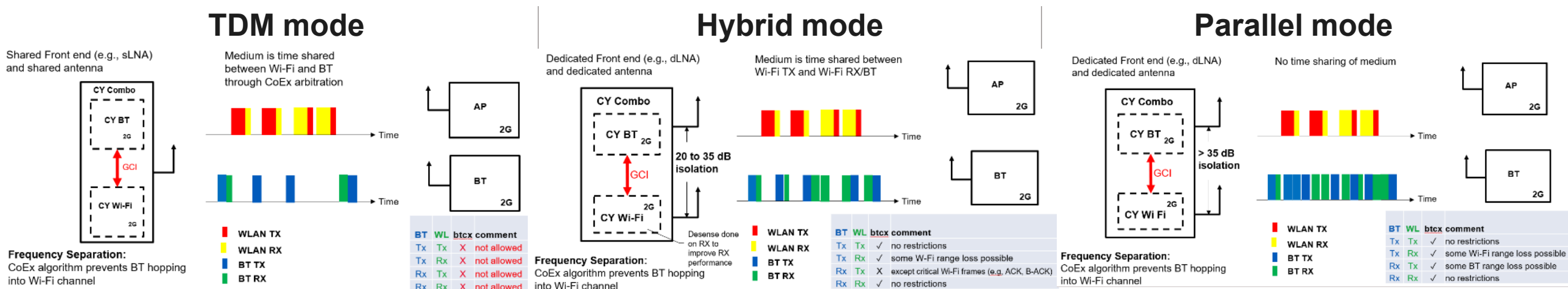
1. Create and program DEEPCRAFT™ Deploy Motion code example using ModusToolbox™
 - i. Example runs on CM55 by default

2. Add Wi-Fi MQTT support
 - i. Update libraries using Library Manager
 - ii. Update CM33_NS project to initiate Wi-Fi connection while CM55 runs ML model
 - iii. Update keys and parameters to connect to MQTT broker
Check parameters with presenter

3. Add VCM support
 - i. Update libraries using Library Manager
 - ii. Update CM33_NS and CM55:
Use safe print leveraging IPC semaphores
Send MQTT message from CM55 using VCM

Efficient Bluetooth® LE audio

AIROC™ CYW55xxx Enhancements- *Smart* CoEx Modes



	TDM mode	Hybrid mode	Parallel mode
Front end (Wi-Fi, BT)	Share Front End (sLNA) Share antenna (x1)	Dedicated Front End (dLNA) Dedicated antenna (x2)	Dedicated Front End (dLNA) Dedicated antenna (x2 or x3)
Isolation	N/A	20 dB - 35dB	> 35 dB
Medium	Time sharing between Wi-Fi and BT through CoEx arbitration	Time sharing between BT Rx and Wi-Fi Tx only	No time sharing of medium
Use case	Small form factor Lower cost	More space Better throughput	No size constraint Best throughput Range is trade-off

AIROC™ Bluetooth® and Bluetooth LE Audio – 5551x

Advantage of using LE Audio over Classic Audio

- **Improved Battery Life:**
 - LE Audio uses Low energy technology for low power consumption
- **Higher Audio Quality at Lower Data Rates:**
 - LC3 codec offers better audio quality at significantly lower data rates
- **Multi Stream Audio:**
 - Connected Unicast independent streams
 - Auracast™ Broadcast independent streams
- **Lower Latency:**
 - LE Audio offers lower latency (delay) compared to Classic Audio for gaming

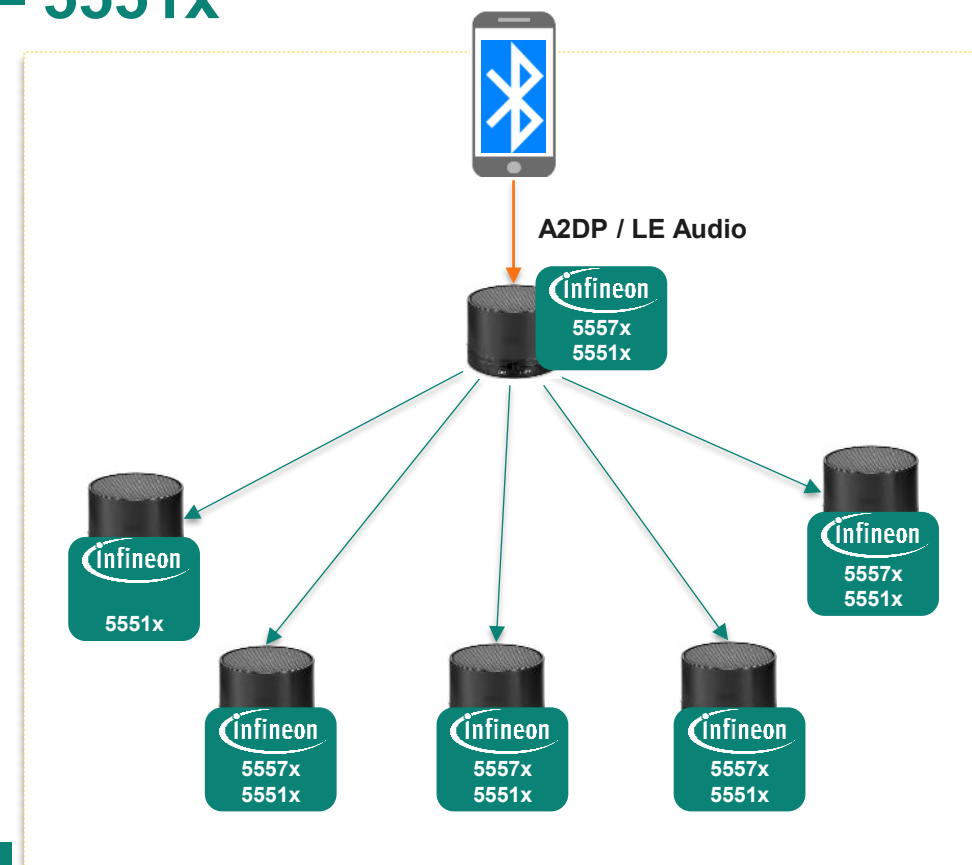
Embedded Mode (5551x):

- 3 High quality music streams and 5 Low Latency gaming streams

Hosted Mode (5557x/5551x):

- 5 High quality music streams and 8 Low Latency gaming streams

* LC3: Low Complexity Communication Codec



From Phone

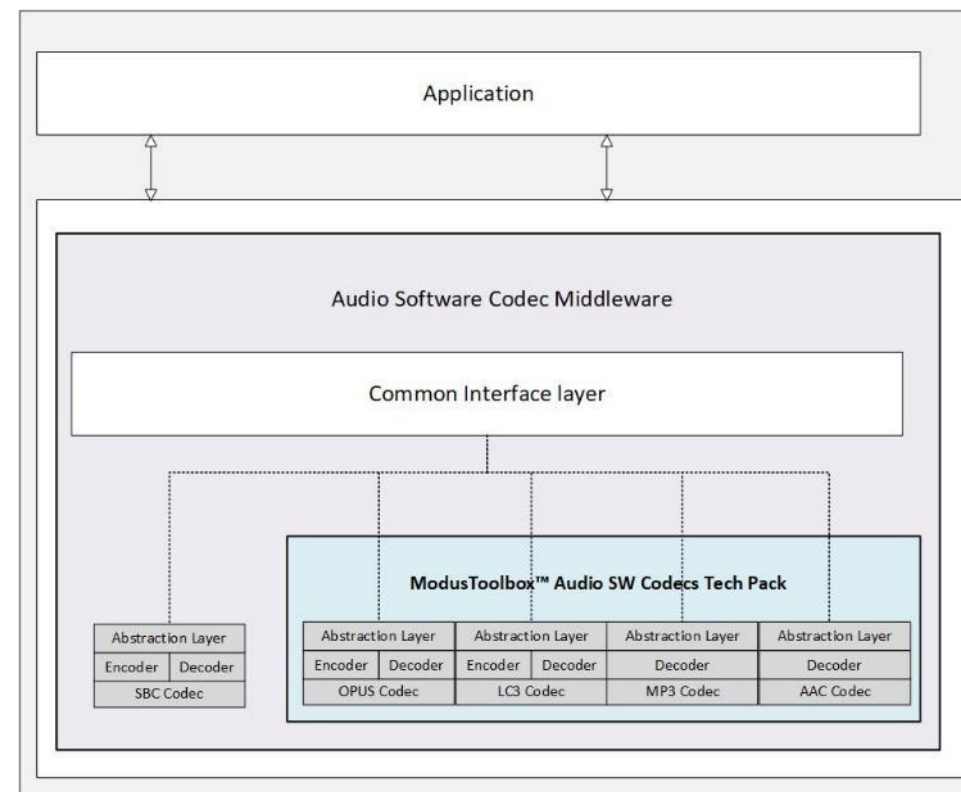
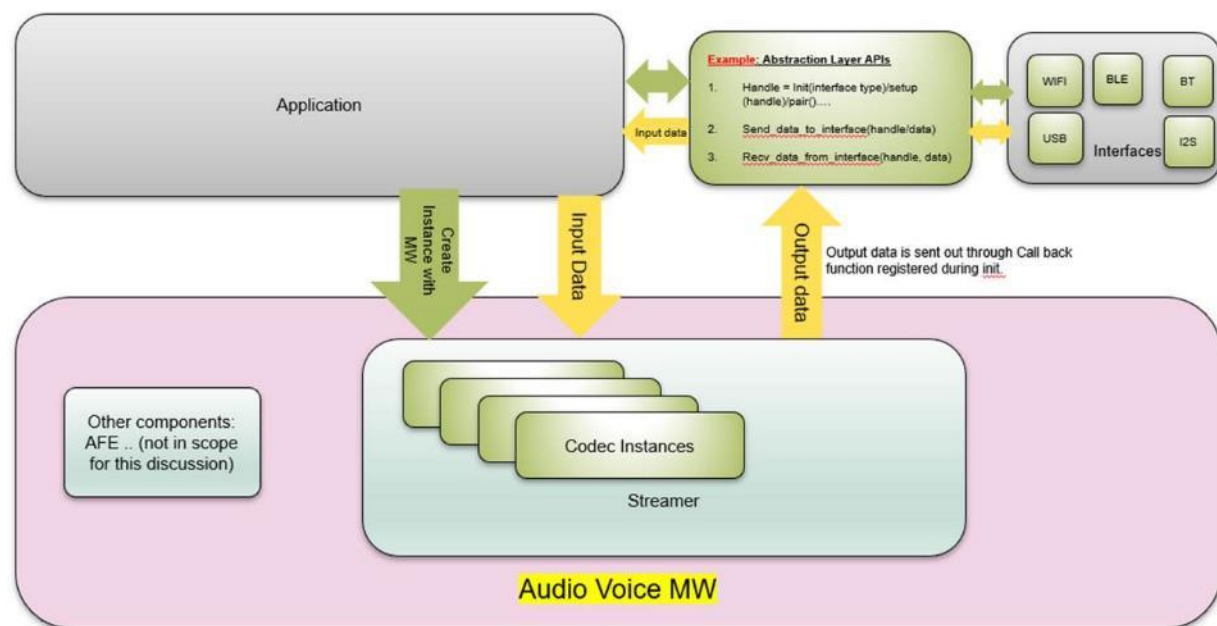
- A2DP Stereo OR
- LE Audio Unicast channel

To Surround Speakers

- 5 LE Audio Unicast channels

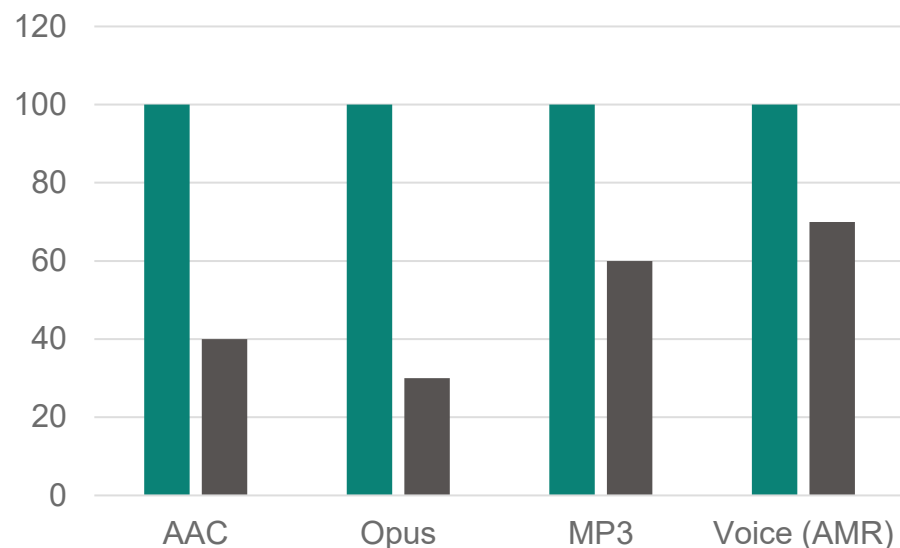
Audio codecs and audio streaming libraries on PSoC™ Edge

- › Provides OPUS, LC3, MP3, SBC, AAC decoders to support music streaming on device
- › Provide encoders to allow offline music storage
- › Provides a common audio workflow to allow interchangeable audio interfaces
 - Switch between amplifier, BT Classic or Wi-Fi
 - Support for Simultaneous interfaces via configuration
- › Support for multiple amplifier drivers



Codec performance using Helium

Typical CPU Reduction vs Non-Helium (Cortex-M4/M7 Baseline)



Source: Arm Helium (MVE) whitepapers, CMSIS-DSP benchmarks, and EEMBC AudioMark™. Results represent typical MCU-based audio workloads (AAC, Opus, MP3, voice codecs) and may vary based on implementation and optimization level.

Performance of audio codec library in PSOC™ Edge (in MCPS)

Codec			Description	GCC v14.2.1	Arm® v6.22	LLVM v19.1.5	IAR v9.2
SBC	Decoder	Mono 48kHz, 16-bit		12.32	7.03	7.24	7.85
	Encoder	Mono 48kHz, 16-bit		9.47	9.88	9.40	9.34
OPUS	Decoder	Mono 16kHz, 16-bit, 32kbps		12.90	9.75	10.05	13.05
	Encoder	Mono 16kHz, 16-bit, 64kbps		21.91	16.91	18.43	28.32
LC3	Decoder	Mono 16kHz, 16-bit, 96kbps		14.41	9.40	9.64	15.66
	Encoder	Mono 16kHz, 16-bit, 96kbps		36.29	20.76	20.73	41.75
MP3	Decoder	Layer 3, 44.1kHz, 128kbps		9.95	8.13	8.15	8.36
AAC	Decoder	AAC only, 128kbps		10.61	8.20	8.33	8.94

– Performance shown in average MCPS using CM55 core at 400MHz

Key Takeaways:

- 2–5x efficiency improvement
- ~60–75% CPU reduction
- Enables always-on audio
- Enables multiple audio streams

Lab 2: Bluetooth® LE Audio

What will you learn?

- How to run a BLE Broadcast sink application
- How the Helium extension of CM55 accelerates audio codecs

Lab 2:

Bluetooth® LE Audio



Follow steps in training manual:

1. Create and program BLE Broadcast sink code example using ModusToolbox™

This example is currently not available on public BSP

- i. Add Audio SW Codec pack library in Library Manager
- ii. Observe performance utilizing Helium (default)

2. Disable Helium extension for LC3 codec

- i. Replace library with custom version
- ii. Observe performance without Helium

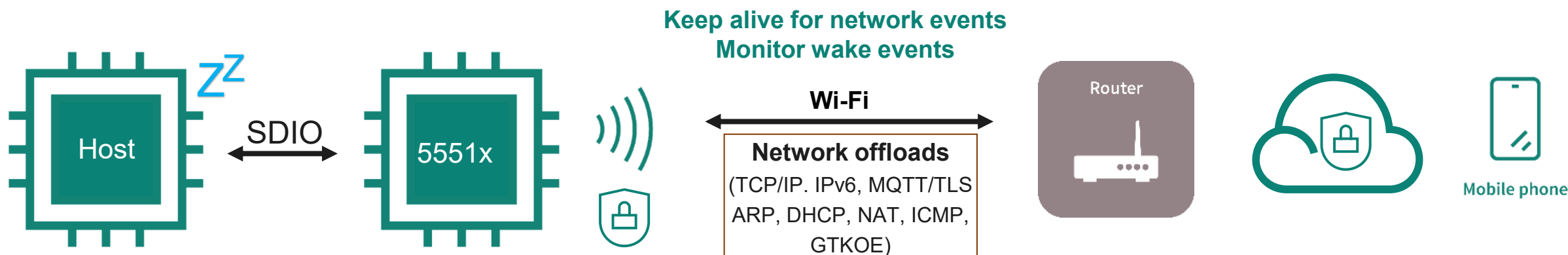
Low-power assistant (LPA)

CYW5551x Optimizes Overall System Power – Network Offloads



Extends battery life for “always connected” Wi-Fi applications

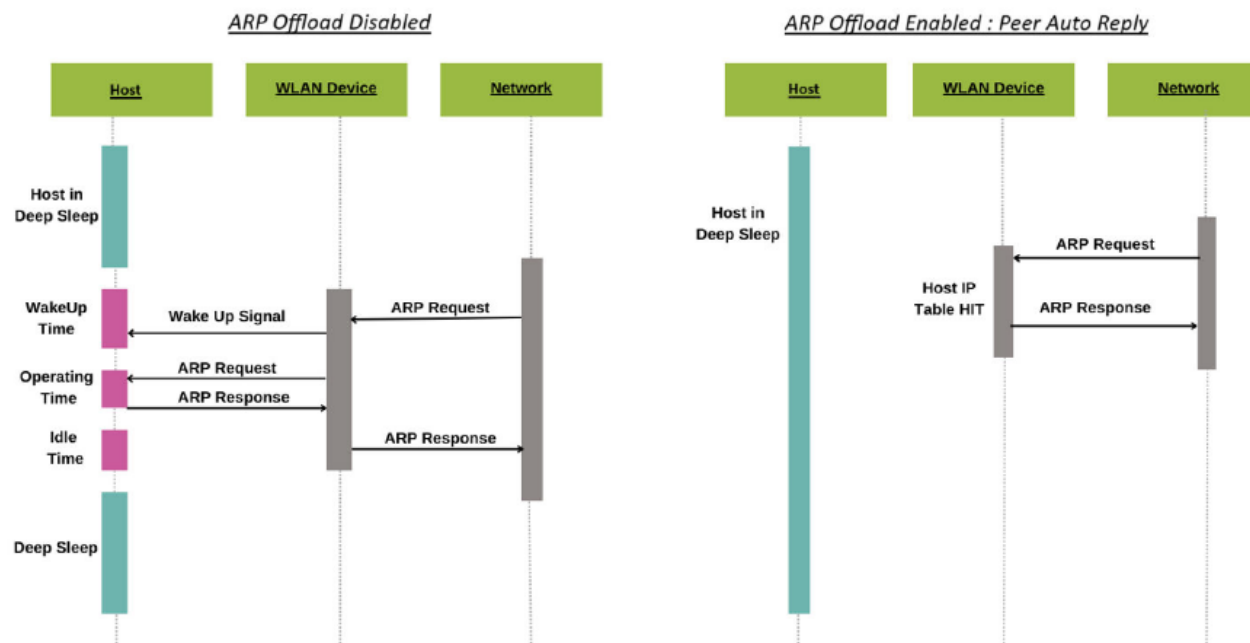
- 5551x maintains network connections
- Performs network “Keep Alive” and “Packet Filter Wake” while host in sleep mode



Event Type	Network Traffic Type
Local Network Traffic	Wake on Wireless Pattern NAT (Network Address Translation)
IP Network Traffic	TCP/IP ARP (Address Resolution Protocol) DHCP (Dynamic Host Configuration Protocol Lease Time Renewal) GTK (Group Temporal Key - key rotation) ICMP (Internet Control Messaging Protocol)
Cloud MQTT Event	MQTT Publish event containing wake request
Matter	IPv6/ Neighbor Discovery mDNS (Mobile Domain Name Service) Simple Service Discovery

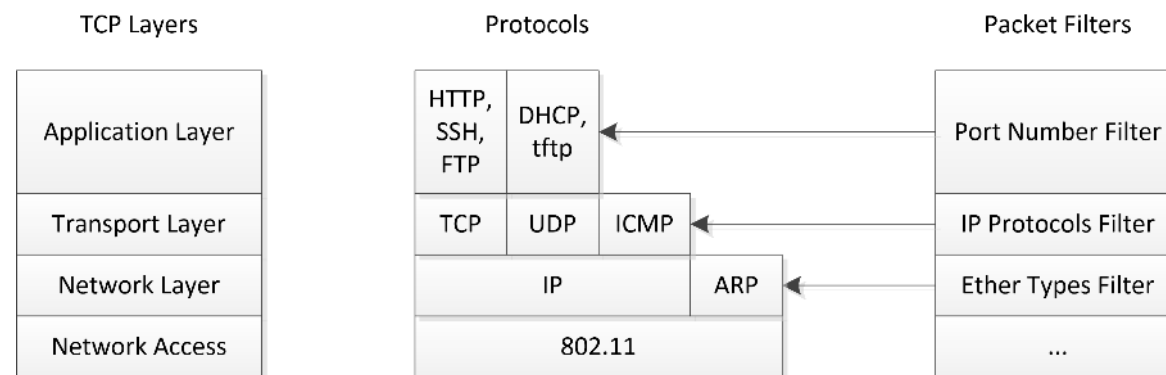
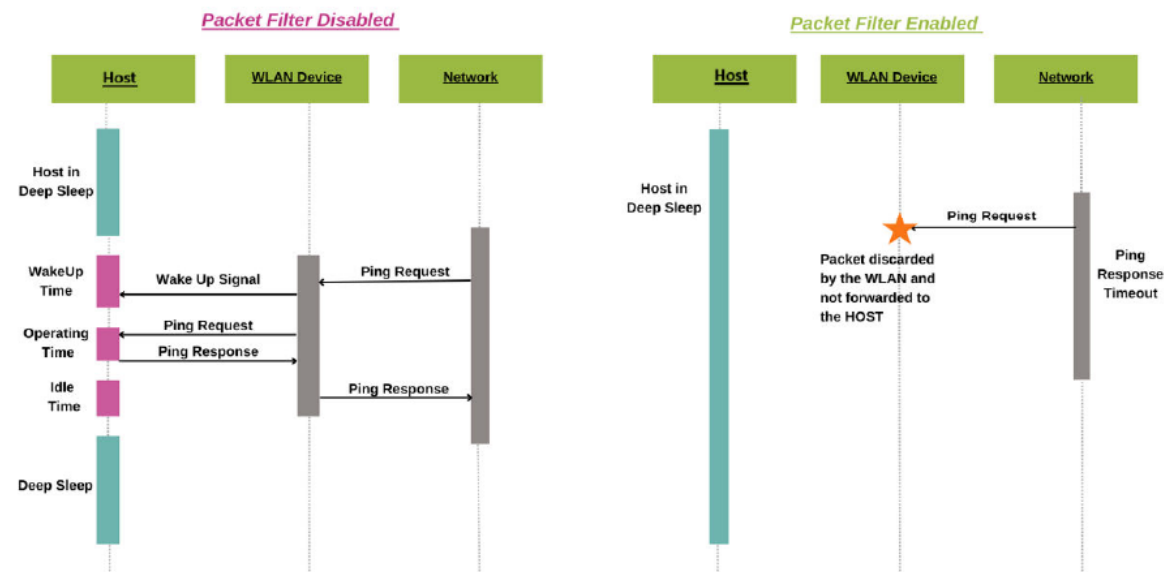
ARP Offload

- **Without LPA:**
 - MCU wakes-up to handle ping requests
- **With LPA:**
 - CYW55513 auto-responds to ARP
- **Benefit:**
 - Maintains network presence while MCU sleeps



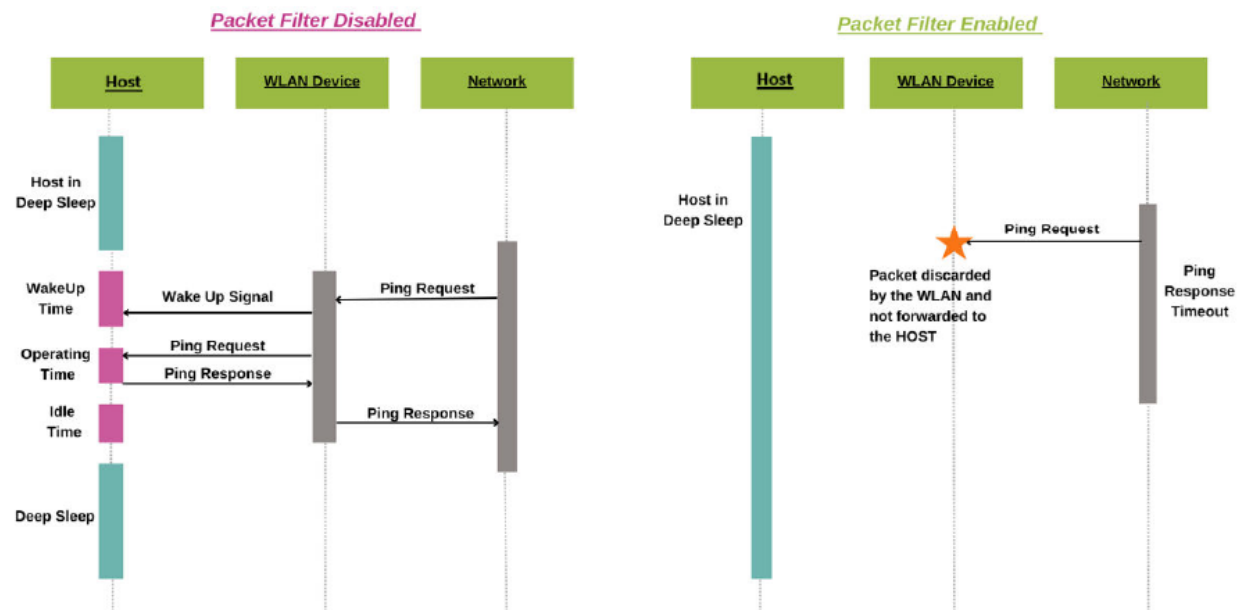
Packet Filter Offload

- **Without LPA:**
 - MCU wakes-up on all packets
- **With LPA:**
 - CYW55513 filters packets and MCU wakes only when needed
- **Benefit:**
 - Reduces unnecessary wakeups → major power savings



TCP Keepalive Offload

- **Without LPA:**
 - MCU sends periodic keepalives
- **With LPA:**
 - CYW55513 maintains connection
- **Benefit:**
 - Persistent connection with minimal host activity



Lab 3: WLAN Low-Power Assistant

What will you learn?

- How to enable LPA
- How to test different LPA offload features

Lab 3:

WLAN Low-Power Assistant



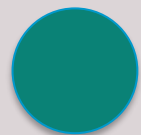
Follow steps in training manual:

1. Create and program WLAN Low-power assistant code example using ModusToolbox™
 - i. Observe device waking up from TCP ping since LPA is disabled (default)
2. Test LPA offload functionality
 - i. Test ARP Offload
 - ii. Test ICMP Offload
 - iii. Configure and test Packet Filter Offload

Other key advantages

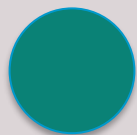
Enterprise Security

Why AIROC™ Wi-Fi/Bluetooth + PSOC™ Edge is a strong fit for enterprise-grade connected AI systems



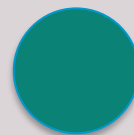
Stronger device security

- WPA3, secure boot support, and modern wireless security features
- Local AI inferencing on PSOC™ Edge reduces cloud data exposure
- Infineon portfolio enables root-of-trust style system designs



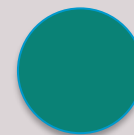
Reliable enterprise connectivity

- Wi-Fi 6/6E improves performance in dense office and industrial deployments
- Bluetooth® supports provisioning, setup, and local service workflows
- Better coexistence and RF robustness help maintain uptime



Lower system power and cost

- Offload functions on AIROC™ reduce host wakeups and MCU workload
- PSOC™ Edge handles AI/ML plus real-time control on one platform
- Shared connectivity architecture lowers BOM and integration complexity



Faster deployment for edge products

- ModusToolbox™ integration speeds software bring-up
- Reference kits shorten prototyping and security evaluation cycles
- Scales across cameras, gateways, access panels, and smart building nodes

Enterprise value: secure onboarding, protected local processing, reliable wireless operation, and efficient always-connected edge AI

Robust & Secure OTA with PSOC™ Edge and AIROC™ (Wi-Fi + BLE)

Flexible, Secure, and Scalable Firmware Updates over Wi-Fi & BLE



- Improved robustness with dual OTA**
 - ✓ High-speed Wi-Fi OTA
 - ✓ Local OTA using BLE
- Faster, independent and more secure multi-blob updates**
 - ✓ Patch only what is needed: App, Connectivity, ML
- Direct XIP from Flash**
 - ✓ Faster updates and more flexible development
- End-to-end Security**
 - ✓ Cloud → transport → device boot → runtime
 - ✓ Trust, protection, reliability, compliance and risk reduction

Faster, safer, and more flexible OTA updates – purpose-built for connected Edge AI systems

Take a minute to share your thoughts!

Your feedback is invaluable to us, and it plays a crucial role in helping us improve and deliver the best training experiences.

Please open your IADT app and complete this session survey

Thank you for taking the time to help us win together!

